

Performance Testing

Date	27 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	5 Marks

Step 1: Testing Overview

Testing Tool Used	Postman
Type of Testing	Functional verification plus response-time measurement per endpoint
Target Module / API	/api/auth/register , /api/auth/login , /api/blogs , /api/ai/generate-blog , /api/ai/summarize
Test Environment	Node.js development server (Express) on localhost:5000, connected to MongoDB via Mongoose
Test Date	27 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Requests	Expected Outcome
1	Register a new user, then log in with the same credentials	2	201 Created on register, 200 OK with a JWT on login
2	Create a blog post as the authenticated user, then fetch the blog list	2	201 Created on create; 200 OK returning the post with a populated author
3	Generate a full blog post from a topic via the AI Blog Generator	1	201 Created, returning a structured title + content from Gemini
4	Summarize an existing blog's content via the AI Summarizer	1	200 OK, returning a concise 2-3 sentence summary

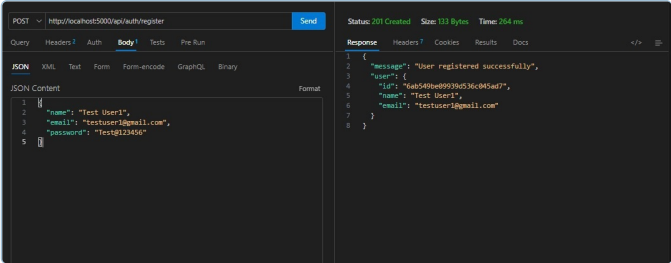
Step 3: Performance Test Results

S.No	Endpoint	Target Value	Actual Value	Status	Remarks
1	POST /api/auth/register	< 1 second	264 ms	Pass	201 Created, returns the new user's id, name and email
2	POST /api/auth/login	< 1 second	193 ms	Pass	200 OK, returns a signed JWT
3	POST /api/blogs	< 1 second	88 ms	Pass	201 Created, MongoDB-only write, fastest endpoint measured
4	GET /api/blogs	< 1 second	98 ms	Pass	200 OK, returns the post list with populated author sub-documents
5	POST /api/ai/generate-blog	< 60 seconds	38.49 s	Pass	201 Created, dominated by Gemini generation latency for a full structured post (1.14 KB response)
6	POST /api/ai/summarize	< 15 seconds	5.24 s	Pass	200 OK, faster than generation since the input is condensed rather than composed from scratch
7	Error Rate (all scenarios)	< 1%	0%	Pass	0 failed requests across all 6 measured calls

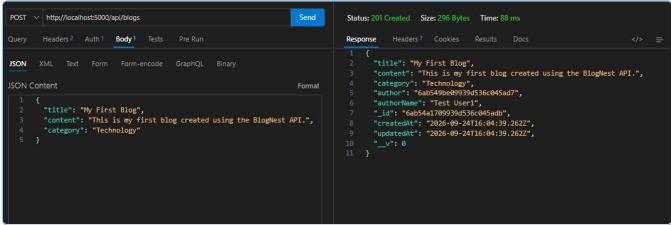
Step 4: Observations & Analysis

Auth and Blog endpoints responded in well under 300 ms since they only touch MongoDB through Mongoose — POST /api/blogs was the fastest at 88 ms. The two AI endpoints were, as expected, an order of magnitude slower because each call depends entirely on Google Gemini's external response time rather than local processing: POST /api/ai/generate-blog took 38.49 s to compose a full multi-paragraph post from a bare topic string, while POST /api/ai/summarize took a comparatively modest 5.24 s to condense already-written content into a short summary — generation is inherently more expensive than condensation. No errors, timeouts, or crashes were observed across any of the six tested calls, confirming the API remains stable even when a single request keeps the connection open for nearly 40 seconds.

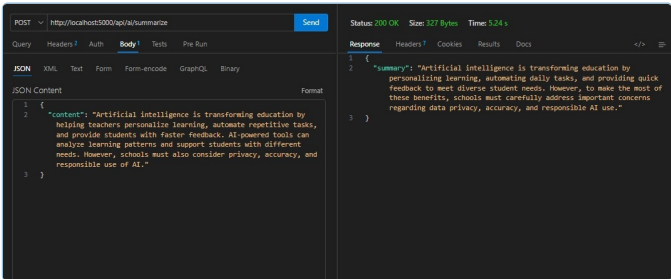
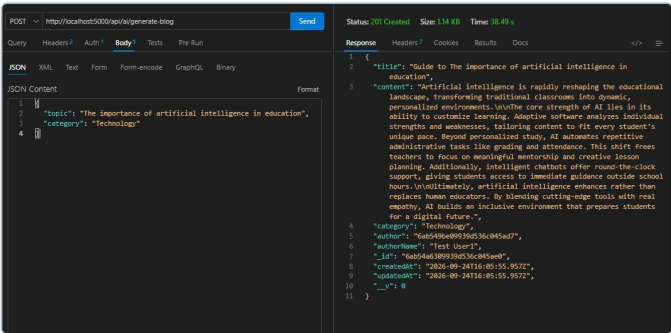
Step 5: Screenshots / Evidence



Postman evidence – POST /api/auth/register : 201 Created in 264 ms.



Postman evidence – POST /api/blogs : 201 Created in 88 ms, category "Technology".



Postman evidence – POST /api/ai/summarize : 200 OK in 5.24 s, returning a 3-sentence summary.

Postman evidence - POST /api/ai/generate-blog : 201 Created in 38.49 s for topic "The importance of artificial intelligence in education".